



Name: _____

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Learning Objectives

- Apply the Power Rule to polynomial and radical functions
- Differentiate e^x , $\ln x$, $\sin x$, and $\cos x$
- Find equations of tangent lines using the derivative
- Use the limit definition of the derivative

Rewrite radicals and rational expressions using exponent notation before differentiating.

1. Differentiate using the Power Rule.

$$f(x) = 5x^4 - 3x^2 + 7x - 2$$

Answer: _____

2. Differentiate. Rewrite radicals as fractional exponents first.

$$f(x) = \sqrt{x} + \frac{1}{x^2}$$

Answer: _____

3. Find the slope of the tangent line to $f(x)$ at $x = 2$.

$$f(x) = x^3 - 4x + 1$$

Answer: _____

4. Differentiate the exponential function.

$$f(x) = 4e^x - 3x^2$$

Answer: _____

5. Differentiate the trigonometric function.

$$f(x) = 3\sin x - 2\cos x$$

Answer: _____

6. Use the definition of the derivative to find $f'(x)$.

$$f(x) = x^2 + 3x, \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Answer: _____



7. Find the equation of the tangent line to $y = f(x)$ at $x = 1$.

$$f(x) = 2x^3 - x + 5$$

Answer: _____

8. Differentiate the natural log function.

$$f(x) = \ln x + 5x^2$$

Answer: _____

9. Find all x -values where $f'(x) = 0$.

$$f(x) = x^3 - 6x^2 + 9x$$

Answer: _____

10. Determine where $f(x)$ is differentiable. State any points of non-differentiability.

$$f(x) = |x - 3|$$

Answer: _____





Problem 2 requires rewriting radicals — common student error. Problem 6 uses the limit definition; connect back to the limit unit. Problem 10 highlights that $|x|$ is non-differentiable at $x = 3$.

Solutions

1. Differentiate using the Power Rule.

$$f(x) = 5x^4 - 3x^2 + 7x - 2$$

→ Apply the Power Rule: $d/dx[x^n] = nx^{n-1}$.

$$\rightarrow d/dx[5x^4] = 20x^3.$$

$$\rightarrow d/dx[-3x^2] = -6x.$$

$$\rightarrow d/dx[7x] = 7. \quad d/dx[-2] = 0.$$

$$\rightarrow f'(x) = 20x^3 - 6x + 7.$$

Answer: $f'(x) = 20x^3 - 6x + 7$

2. Differentiate. Rewrite radicals as fractional exponents first.

$$f(x) = \sqrt{x} + \frac{1}{x^2}$$

→ Rewrite: $f(x) = x^{1/2} + x^{-2}$.

$$\rightarrow f'(x) = (1/2)x^{-1/2} + (-2)x^{-3}.$$

$$\rightarrow = 1/(2\sqrt{x}) - 2/x^3.$$

Answer: $f'(x) = \frac{1}{2\sqrt{x}} - \frac{2}{x^3}$

3. Find the slope of the tangent line to $f(x)$ at $x = 2$.

$$f(x) = x^3 - 4x + 1$$

$$\rightarrow f'(x) = 3x^2 - 4.$$

$$\rightarrow f'(2) = 3(4) - 4 = 12 - 4 = 8.$$

→ The slope of the tangent at $x = 2$ is 8.

Answer: $f'(2) = 3(2)^2 - 4 = 8$

4. Differentiate the exponential function.

$$f(x) = 4e^x - 3x^2$$

$$\rightarrow d/dx[e^x] = e^x.$$

$$\rightarrow f'(x) = 4e^x - 6x.$$

Answer: $f'(x) = 4e^x - 6x$

5. Differentiate the trigonometric function.

$$f(x) = 3\sin x - 2\cos x$$

$$\rightarrow d/dx[\sin x] = \cos x. \quad d/dx[\cos x] = -\sin x.$$

$$\rightarrow f'(x) = 3 \cos x - 2(-\sin x) = 3 \cos x + 2 \sin x.$$

Answer: $f'(x) = 3\cos x + 2\sin x$



6. Use the definition of the derivative to find $f'(x)$.

$$f(x) = x^2 + 3x, \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\rightarrow f(x+h) = (x+h)^2 + 3(x+h) = x^2 + 2xh + h^2 + 3x + 3h.$$

$$\rightarrow f(x+h) - f(x) = 2xh + h^2 + 3h = h(2x + h + 3).$$

$$\rightarrow \text{Divide by } h: 2x + h + 3.$$

$$\rightarrow \text{As } h \rightarrow 0: f'(x) = 2x + 3.$$

Answer: $f'(x) = 2x + 3$

7. Find the equation of the tangent line to $y = f(x)$ at $x = 1$.

$$f(x) = 2x^3 - x + 5$$

$$\rightarrow f'(x) = 6x^2 - 1. \quad f'(1) = 6 - 1 = 5 \text{ (slope).}$$

$$\rightarrow f(1) = 2 - 1 + 5 = 6. \text{ Point: } (1, 6).$$

$$\rightarrow \text{Tangent: } y - 6 = 5(x - 1) \rightarrow y = 5x + 1.$$

Answer: $y = 5x + 1$

8. Differentiate the natural log function.

$$f(x) = \ln x + 5x^2$$

$$\rightarrow d/dx[\ln x] = 1/x.$$

$$\rightarrow f'(x) = 1/x + 10x.$$

Answer: $f'(x) = \frac{1}{x} + 10x$

9. Find all x -values where $f'(x) = 0$.

$$f(x) = x^3 - 6x^2 + 9x$$

$$\rightarrow f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3).$$

$$\rightarrow \text{Factor: } 3(x-1)(x-3).$$

$$\rightarrow f'(x) = 0 \text{ when } x = 1 \text{ or } x = 3.$$

Answer: $x = 1$ and $x = 3$

10. Determine where $f(x)$ is differentiable. State any points of non-differentiability.

$$f(x) = |x - 3|$$

$$\rightarrow f(x) = x - 3 \text{ for } x \geq 3 \text{ (derivative = 1).}$$

$$\rightarrow f(x) = -(x - 3) \text{ for } x < 3 \text{ (derivative = -1).}$$

$$\rightarrow \text{Left and right derivatives differ at } x = 3 \text{ — corner point.}$$

$$\rightarrow f \text{ is differentiable everywhere except } x = 3.$$

Answer: Not differentiable at $x = 3$

